power

J107 B6B-ZR-3.4(LF)(SN)

Q101 SS8050 · Q102 SS8550

U101 L78L05_SOT89 · U102 L78L05_SOT89 · U103 XC6206P332MR-G · U104 NJW4191M

D101 LED(RED) p8V · D103 LED(RED) p5V · D106 LED(RED) p3v3 · D107 1N5819WS · D108 LED(RED) n8V

Because the NJW4191's output contains significant switching noise, a ripple filter is required at its output.

R110 10k / R111 3.3k set the PowerGood window together with the firmware constants PGOOD_DIV_LO_MV / PGOOD_DIV_HI_MV in main.c. The firmware does not know the divider ratio at run time: it compares the measured voltage against mV values pre-computed from 3.3/13.3 = 0.24812. Changing these resistors without recomputing both constants shifts the trip points silently — no error is raised. 1% parts are required. The firmware window is 8.0–12.0 V (nominal): worst-case unit tolerance of +−2.6% (+−1.5% from the 1% divider) can move the trips to 8.21 V / 11.69 V, leaving 0.19 V of margin to the specified 8.4–11.5 V operating range. With 5% parts (divider +−7.5%) warnings can fire inside that range. At 12 V the divider gives 2.977 V, 0.3 V below VDDA. Raising the ratio saturates the ADC and the upper trip stops working.

BOOT0 is pulled low by a 10k to ensure boot from the internal Flash memory.

U105 STM32F030C8T6

J101 Conn_ST_SDC14 · J102 FootSW B3B-PH-K-S · J103 PH-2x10SG · J104 MJ-8435 · J105 PH-2x4SG North-West

SW101 SW_SPST (Reset) · SW102 KSD42 · SW103 SW_SPST (MIDI-Learn)

FB101 BLM18PG121SN1D

D109 BAV99 · D111 LED(RED) MIDI_IN · D112 LED(RED) MIDI_OUT · D113 LED(RED) PG · D114 LED(RED) CLK · D116 BAV99 · D117 BAV99 · D118 1N5819WS

U106 M24C64-RMN6TP · U107 SN74HCT04DR · U107G SN74HCT04DR · U109 SN74LVC1G66DCKR · U110 SN74LVC1G66DCKR

U107 must be HCT, not HC. It runs on +5 V, but 3.3 V MCU GPIO or TIM drive its inputs directly. HC thresholds scale with VCC and 3.3 V does not reach them at 5 V. HC and HCT are pin- and function-compatible, so the substitution looks harmless on the schematic but will not work.

J104's function is determined by the state of PB5 (GPIO_OUTPUT).
PB5 = H: EXP-SW (latch) Mode
PB5 = L: MIDI-OUT Mode
(MIDI TRS Type A compliant)

MIDI OUT / EXP-SW IN

v1.1: Replaced the BJT open-collector circuit with two TTL-compatible inverters in series (U107A, U107B).
In EXP-SW (latch) Mode, D118 is required to prevent U107B from sourcing current.

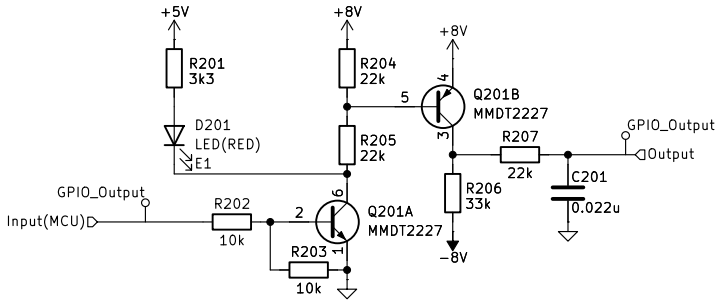
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File: anthracite-bottom.kicad_sch

Title: Anthracite Schematic (Bottom)

Size: A3	Date: 2026-11-11	**Rev: v1.1**
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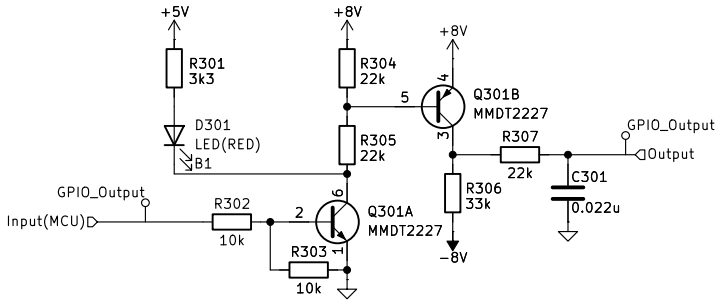
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Title: Anthracite Schematic (Bottom)

Size: A4	Date: 2026-11-11	**Rev: v1.1**
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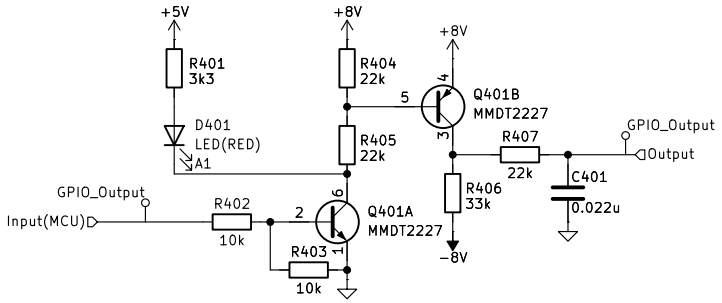
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Title: Anthracite Schematic (Bottom)

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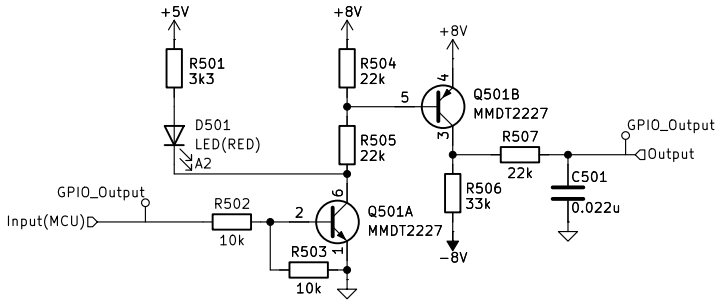
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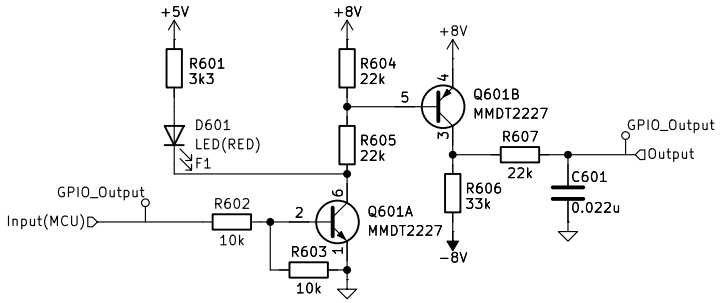
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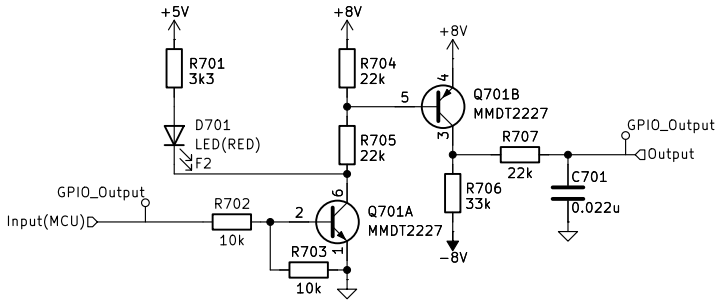
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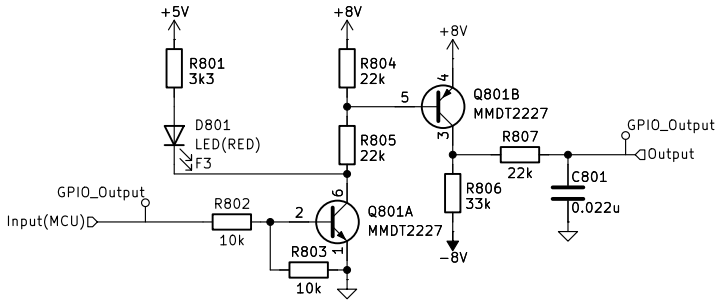
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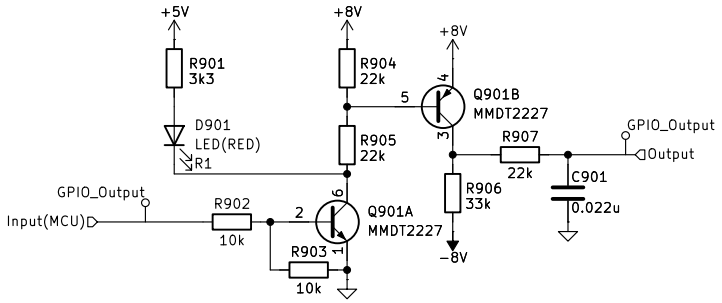
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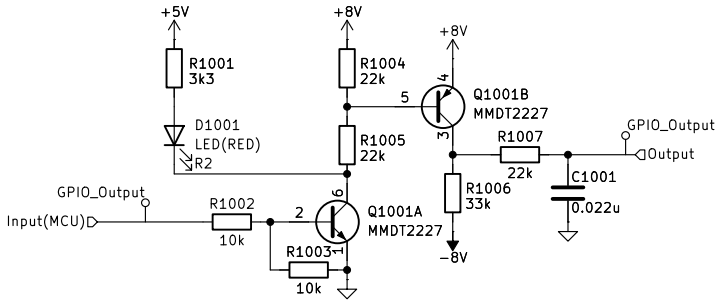
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